

VIJAYA DURGA REDDY PADALA

AI FULL STACK ENGINEER

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PROFESSIONAL SUMMARY

Senior AI Full Stack Engineer with 5+ years of experience building and supporting scalable enterprise applications across **Generative AI, full-stack development, backend engineering, cloud computing, microservices, and API development**. Strong hands-on experience with **Python, Java, JavaScript, TypeScript, React.js, Node.js, Spring Boot, FastAPI, REST, GraphQL, AWS, and Azure**. Experienced in integrating **LLMs, OpenAI APIs, RAG, embeddings, semantic search, AI agents, and GenAI capabilities** into application workflows. Skilled in developing secure APIs, event-driven services, cloud-native applications, automated workflows, CI/CD pipelines, and production-ready services using **Docker, Kubernetes, Terraform, GitHub Actions, and Jenkins**. Experienced across frontend and backend development, databases, cloud infrastructure, testing, application security, and Agile software development.

TECHNICAL SKILLS

AI / Generative AI: LLMs, Generative AI, OpenAI APIs, GPT Models, RAG, Embeddings, Semantic Search, AI Agents, Prompt Engineering, NLP/NLU, Machine Learning, TensorFlow, NumPy, Model Evaluation

Programming Languages: Python, Java, JavaScript, TypeScript, SQL

Frontend Development: React.js, Angular, React Native, Redux, HTML, Tailwind CSS, jQuery, AJAX

Backend Development & APIs: Node.js, FastAPI, Java, Spring Boot, Spring MVC, Spring Security, Hibernate, Express.js, NestJS, REST APIs, GraphQL, Microservices

Cloud & DevOps: AWS, Azure, EC2, S3, Lambda, DynamoDB, Azure Functions, Azure App Service, Azure SQL, Azure Storage Queues, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD

Databases: PostgreSQL, MongoDB, MySQL, Oracle, Firebase, Neo4j, Azure SQL

Event-Driven Architecture & Integration: Kafka, Azure Storage Queues, Azure Functions, Webhooks, Data Synchronization, Third-Party API Integrations

Testing & Development Tools: Git, Bitbucket, JIRA, Postman, JUnit, Selenium, Cypress, Playwright, Jest, Maven

PROFESSIONAL EXPERIENCE

SENIOR SOFTWARE ENGINEER II

Temple University Health System

April 2026 – Present

Philadelphia, PA

- Lead development of scalable AI-enabled full-stack enterprise applications using **React.js, Python, Typescript, JavaScript and Azure**.
- Developed responsive patient-facing web experiences using **React.js and TypeScript** and cross-platform iOS/Android functionality using **React Native**, with reusable components and shared application patterns.
- Developed backend APIs and application services using **Node.js/Express/NestJS and Python/FastAPI** to support patient workflows, educational content, notifications and integrated with existing **Java/Spring Boot enterprise services**.
- Integrate AI-powered agent functionality into the application to provide contextual assistance and guided user experiences, using **LLM APIs, prompt orchestration, structured responses, and secure backend-mediated model access**.
- Implement **RAG, embeddings, semantic search, and vector-based retrieval** for enterprise content and knowledge workflows.
- Integrated the application with **Epic/MyChart and FHIR-based workflows**, connecting React/React Native clients to existing **Java/Spring Boot** enterprise services and secured healthcare APIs.
- Implemented secure authentication and authorization using **OAuth 2.0, OpenID Connect/SMART on FHIR, PKCE, token validation, and protected API access** for patient-facing workflows.
- Integrated enterprise **CMS/content APIs** to deliver educational material, care guidance, and age- or journey-specific content across maternal, newborn, and pediatric experiences.
- Implemented mobile notification workflows using **Firebase Cloud Messaging and Apple Push Notification Service** for appointment, engagement, and care-related notifications.
- Deployed and supported production services using **Azure App Service, Azure Functions, Azure SQL, Key Vault, Application Insights, Storage, Terraform, and CI/CD pipelines**.
- Established automated testing across frontend, mobile, APIs, and integration flows using **Jest, Cypress, Playwright, Selenium, and API testing tools**, while leading code reviews, architecture discussions, sprint planning, and release-readiness activities.

SENIOR SOFTWARE ENGINEER

Capital One

March 2025 – March 2026

Richmond, VA

- Developed and maintained backend microservices using **Java, Spring Boot, REST/GraphQL APIs, and PostgreSQL** to support online payments, transaction processing, customer financial workflows, and external service integrations.
- Designed and implemented secure **REST and GraphQL APIs** for payment initiation, transaction-status processing, customer data exchange, and downstream financial-service integrations.

- Integrated the platform with external **fintech providers and communication services** using REST APIs, webhooks, and asynchronous processing patterns.
- Built automated data-synchronization workflows to reconcile transaction data, process status updates, and maintain consistency across internal and external systems.
- Designed and optimized **PostgreSQL schemas, queries, indexes, and transaction-handling logic** for high-volume financial data and payment-processing workflows.
- Developed resilient integration patterns using validation, retry handling, idempotency, error handling, and logging to improve reliability of payment and third-party service workflows.
- Containerized services using **Docker** and contributed to automated build, test, and deployment workflows using **GitHub Actions CI/CD**.
- Monitored, debugged, and resolved production issues across APIs, database workflows, external integrations, and transaction-processing services.
- Participated in architecture discussions, code reviews, testing, Agile ceremonies, and cross-functional collaboration with product, QA, platform, and integration teams.

JR. SOFTWARE ENGINEER

October 2021 – August 2023

Cognizant Technology Solutions

Chennai, India

- Worked on customer-facing banking and insurance applications using **Java, Spring Boot, Hibernate, AngularJS, REST APIs, and Oracle**.
- Developed and maintained features related to **customer transactions, rewards, account activity, and insurance services** across web and backend applications.
- Integrated the application with **third-party payment and service gateways**, handling request/response flows, validation, error scenarios, and status updates.
- Worked on secure login and API access using **Spring Security, OAuth, and role-based authorization** for customer and internal workflows.
- Improved Oracle and Hibernate queries used for transaction history, reducing data retrieval time by **25%**.
- Supported frontend enhancements using **AngularJS, JavaScript, AJAX, HTML, and CSS**, and connected UI screens with Spring Boot backend services.
- Investigated production issues across UI, APIs, database queries, and external integrations, working with QA and other development teams to resolve defects.

PROGRAM ANALYST

February 2021 – September 2021

Cognizant Technology Solutions

Chennai, India

- Supported day-to-day development and maintenance of a **product management platform** covering product catalogs, inventory workflows, and website content updates.
- Developed and updated reusable frontend components using **React.js, JavaScript, HTML5, CSS3, and Redux** while learning component-based application development.
- Connected frontend screens with **Java/Spring Boot REST APIs** to retrieve and update product, catalog, and inventory information.
- Worked on **catalog and inventory workflows**, helping ensure product information and availability changes were reflected correctly across the web application.
- Supported **real-time or near-real-time UI updates** for product and inventory changes using API-driven data refresh and asynchronous frontend processing.
- Participated in **code reviews, defect fixes, testing, and production-support activities**, working with senior developers to understand coding standards and application architecture.
- Completed hands-on learning in **full-stack development and introductory AI/ML concepts**, applying those skills through internal exercises and project-based development where applicable.

AI & FULL STACK PROJECTS

MANGO LEAF DISEASE DETECTION

Machine Learning Research

- Developed a machine learning model using **Python, NumPy, TensorFlow, image preprocessing, feature extraction, and classification**.
- Achieved **92% accuracy** in mango leaf disease detection and published the research in **IJERT**.

BUG TRACKING SYSTEM

Full Stack Application

- Built a full-stack bug tracking application using **React.js and Spring Boot**.
- Developed frontend functionality and backend REST services for application workflows.
- Improved dashboard latency by **20%** through application optimization and caching.

EDUCATION

MASTER OF SCIENCE IN COMPUTER AND INFORMATION SCIENCE

September 2023 – December 2024

Sacred Heart University

Fairfield, CT

CGPA: **3.98/4.0** | Upsilon Pi Epsilon (UPE) Honor Society

BACHELOR OF TECHNOLOGY IN ELECTRONICS AND COMMUNICATION

April 2017 – May 2021

Saveetha School of Engineering

Chennai, India

CGPA: **7.89/10**

